# 第 7 組 統計實習 Summary

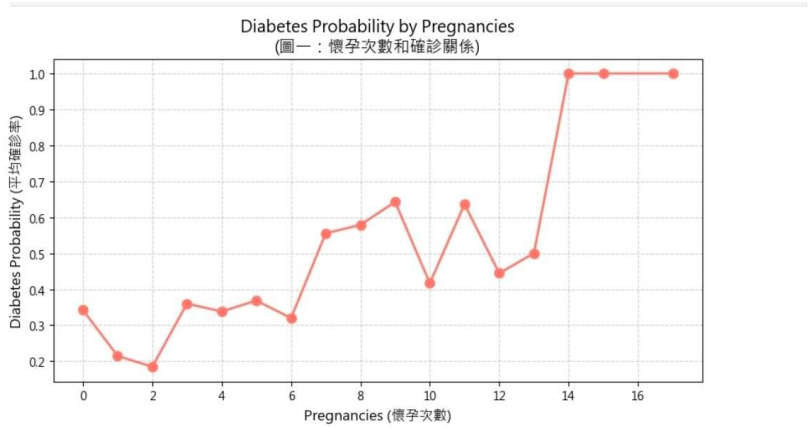



Figure 1: Superficial Positive Correlation. The average diagnosis rate of diabetes generally increases with the number of pregnancies, especially reaching 100% after 14 or more pregnancies. Without in-depth analysis, it is easy to be misled into thinking that "multiple pregnancies directly lead to diabetes."

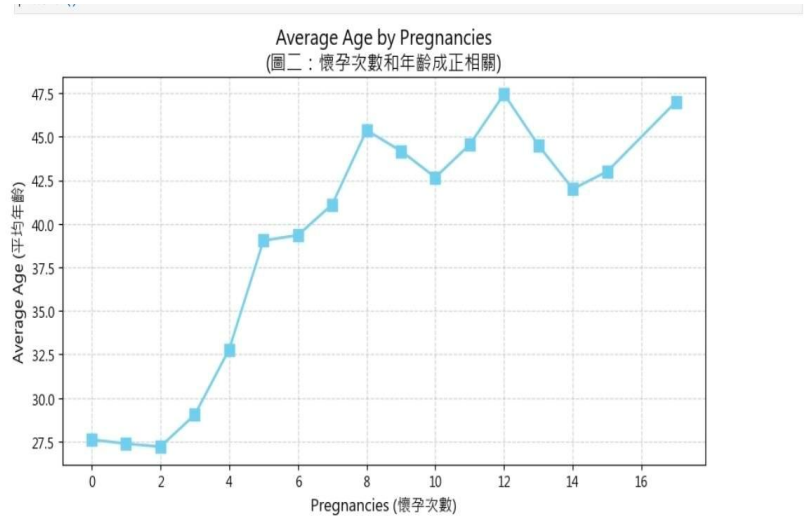
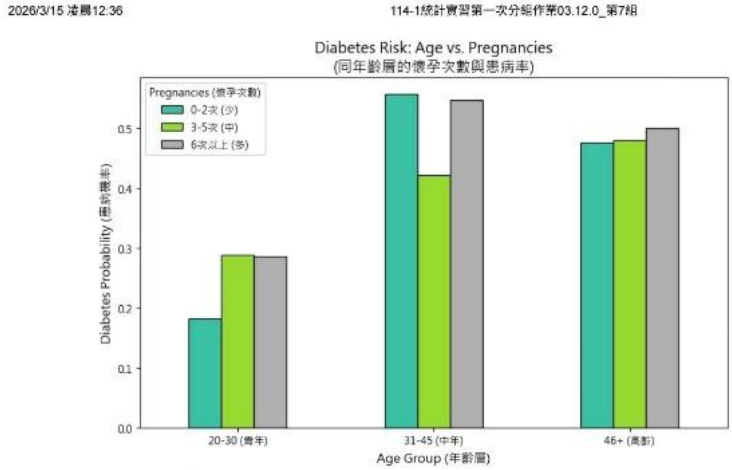



Figure 2: Discovery of a confounding variable (age) Number of pregnancies is highly correlated with average age. For example, women who have had 0-2 pregnancies have an average age of approximately 27.5 years, while those who have had 12 or more pregnancies have an average age closer to 45-47 years. This suggests that "multiple pregnancies" generally implies "older age."




Out[31]:

| Preg_Group | 0-2次 (少) | 3-5次 (中) | 6次以上 (多) |
|---|---|---|---|
| Age_Group | | | |
| 20-30 (青年) | 0.182456 | 0.288288 | 0.285714 |
| 31-45 (中年) | 0.558140 | 0.421875 | 0.547619 |
| 46+ (高齡) | 0.476190 | 0.480000 | 0.500000 |

Figure 3: The Truth After Grouping (Control Variables)Within the same age group: Within the 46+ age group (advanced maternal age group), regardless of the number of pregnancies (few, moderate, or multiple), the diagnostic rate was extremely similar.Across age groups: As the age group increased, the diagnostic rate showed a significant step-like up. This establishes that age is the main driving force.